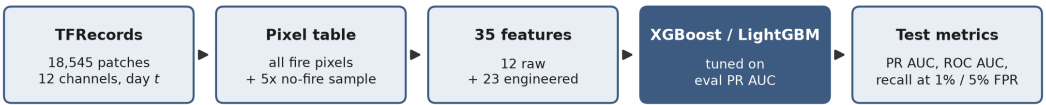
Next Day Wildfire Spread Prediction

Phase 1: per-pixel boosted trees with engineered spatial features

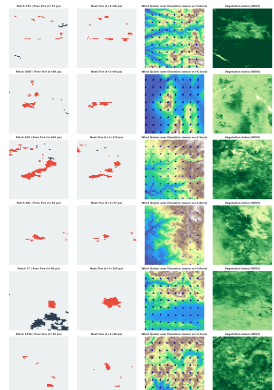
Harshil Sushant Priyanshu Sachin Piyush

AML course project CS & AI Department, Rishihood University



Problem: which pixels burn tomorrow?

- Input: 12 channels on day t for a 64×64 km patch at 1 km (weather, terrain, vegetation, population, today's fire)
- Output: fire mask on day $t{+}1$, one probability per pixel
- Only 1.1% of labelled pixels burn (90:1); cloud or smoke pixels (-1) are ignored
- An all-zero model scores over 98.7% accuracy, so we report **PR AUC**: precision and recall on the rare fire class, over all thresholds



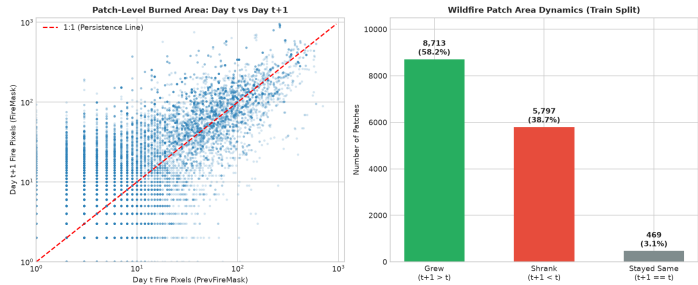
Related work at a glance

- Other spread models use different data and metrics, so their scores are context only (white rows)
- Huot et al. 2022 built our dataset: their pixel models reach 0.225 PR AUC, the whole-patch CNN 0.284
- FireCast: 91% recall but only 3.6% precision. This is why we report PR AUC, not accuracy

Paper	Method	Data	Metric	Result
Hodges & Lattimer 2019	CNN (DCIGN)	simulated fires	F vs. simulation	0.93
Radke et al. 2019	CNN + weather	Rocky Mts., 30 m	precision / recall	3.6% / 91.1%
Gerard et al. 2023	U-Net / UTAE	VIIRS, 375 m	AP	0.341 / 0.372
Huot et al. 2022	persistence	NDWS, 1 km	PR AUC	0.115
Huot et al. 2022	LR + 8 neighbours	NDWS	PR AUC	0.198
Huot et al. 2022	RF + 8 neighbours	NDWS	PR AUC	0.225
Huot et al. 2022	CNN autoencoder	NDWS	PR AUC	0.284

The dataset and why accuracy is useless

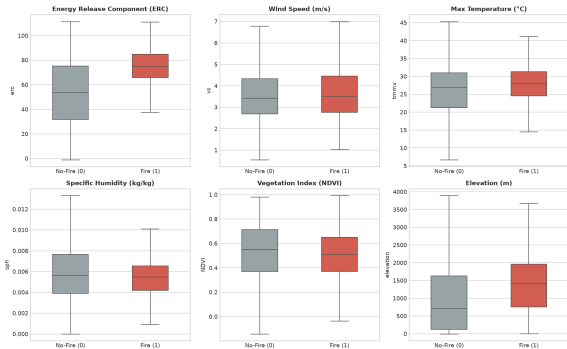
- 18 545 patches of US wildfires (2012–2020), 64×64 km at 1 km, 12 channels on day $t \rightarrow$ fire mask on day $t{+}1$
- Split by calendar week 8:1:1 with a 1-day buffer: 14 979 / 1 877 / 1 689 patches
- Next-day fire = 1.1% of labelled pixels (90:1); an all-zero model scores >98.7% accuracy $\rightarrow$ we report PR AUC



58% of fires grow from day t to $t{+}1$, 39% shrink.

What the raw data looks like

- Export artefacts: wind direction from $-505\,870^\circ$ to $37\,736^\circ$, temperatures from -445 K to $1\,230$ K, negative rain in 9.8% of pixels $\rightarrow$ clipped or wrapped
- ERC (75 vs 54) and elevation ($1\,414$ vs 709 m) separate fire from no-fire; wind speed and humidity barely differ
- Day-t fire has $|\rho| \leq 0.09$ with every raw channel $\rightarrow$ the signal is spatial, not in a pixel's own weather



Method: from TFRecords to a per-pixel model

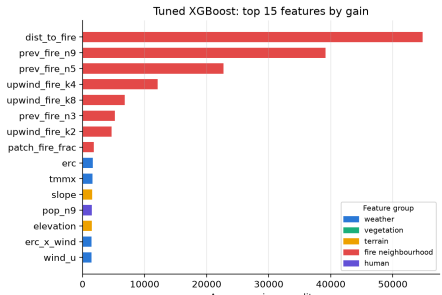


- TFRecords parsed without TensorFlow into $(N, 12, 64, 64)$ arrays
- Artefacts clipped to fixed physical bounds; wind direction wrapped (θ mod 360)
- Patches are stored south-up; we verified this from slope greenness and downwind spread

- Training table: every fire pixel + 5:1 no-fire sample = 3.95 M rows; `scale_pos_weight` = 5
- 20 random configurations per model, early stopping on eval, best by eval PR AUC
- Decision threshold: best F1 on eval, applied unchanged to test

Features: 12 raw + 23 engineered, in five groups

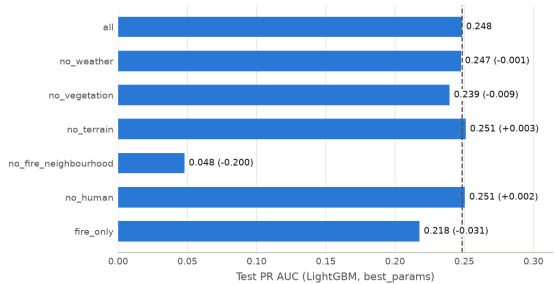
Group	#	Engineered examples
Fire neighbourhood	10	fire density (3/5/9 px), distance, fire upwind
Weather	14	wind u/v, VPD, min. rel. humidity, ERC $\times$ wind
Vegetation	4	local NDVI mean, std, anomaly
Terrain	5	slope, aspect, relative elevation
Human	2	population, 9×9 mean



- `dist_to_fire`: univariate AUC 0.866 vs 0.593 for the raw fire mask
- Most XGBoost gain is in fire features: `dist_to_fire` 32%, `prev_fire_n9` 23%, upwind 13.5%

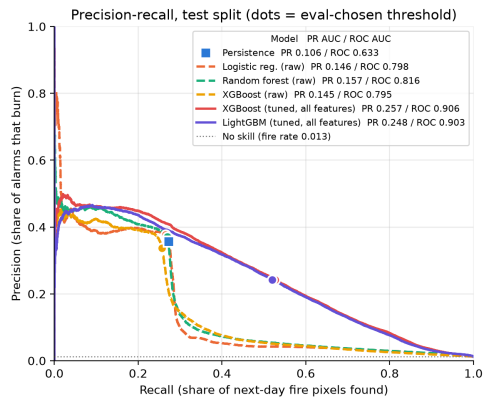
From EDA to features, and what each group is worth

- 40% of new fire is more than 5 km from day-t fire $\rightarrow$ added `upwind_fire_k8` (ranks 5th of 35 by XGBoost gain)
- Ablation: dropping fire-neighbourhood features drops PR AUC 0.248 $\rightarrow$ 0.048; vegetation is the most useful environmental group (-0.009)
- Fire features alone: 0.218; environment adds $+0.031$ on top



Engineered features beat every baseline

- Tuned XGBoost: **PR AUC 0.257**, vs 0.145–0.157 for raw-channel models and 0.106 for persistence (LightGBM 0.248)
- Finds 39% of next-day fire at 1% FPR and 68% at 5% FPR; highest F1 of all models (0.332)
- Above Huot et al.'s neighbour RF (0.225), below their CNN (0.284)



All numbers on the test split; F1 at the threshold chosen on eval.

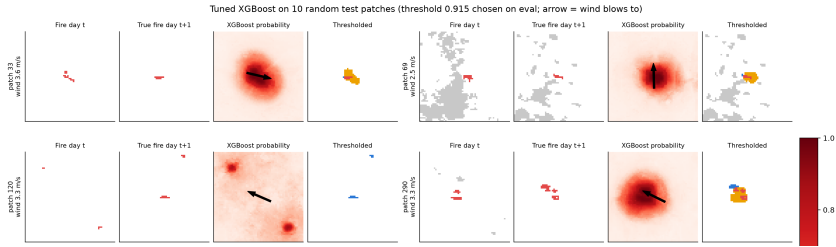
Where we stand

- Our XGBoost on 35 features (12 raw + 23 engineered): 0.257 PR AUC on the same test split (95% interval over fire events: 0.19–0.33)
- Between Huot's random forest (0.225) and CNN (0.284), both inside our interval; recent U-Nets reach about 0.35 (Esparza et al. 2026 preprint)
- Tuning moves PR AUC by about 0.01, spatial features by about 0.20. Phase 2: a CNN / U-Net on the whole patch



What the model learned

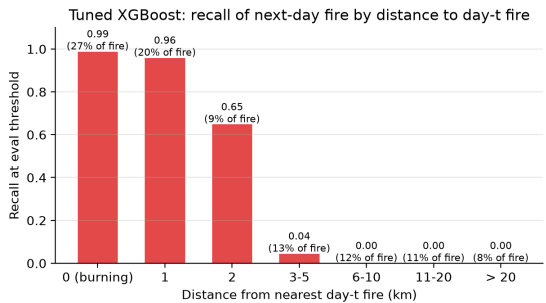
- Predicts a halo around today's fire; fire-neighbourhood features dominate (SHAP 2.29 vs 0.40 for weather)
- Physically sensible signs: near fire, dry air, strong wind, fire upwind → higher risk (ERC is the exception)
- Pushes risk downwind; train data agrees: downwind/upwind ratio grows from 1.135 (calm) to 3.100 (≥ 8 m/s)



Test patches. Red = fire / true positive, orange = false positive, blue = missed; arrow = wind blows to.

Where it fails → Phase 2

- False alarms around dying fires: 88.8% within 2 km of today's fire; only 35.3% of burning pixels still burn tomorrow
- Misses long-range spread: recall 0.988 at the fire, 0.648 at 2 km, 0 beyond 5 km
- Fixed-radius halo, no fire-front shape → Phase 2: CNN / U-Net that sees the whole patch



Conclusion

- A per-pixel model with 23 engineered spatial features reaches test PR AUC **0.257** (95% interval over fire events 0.19–0.33), against 0.106 for persistence
- Huot et al.'s RF (0.225) and CNN (0.284) both lie inside our interval: we land between them, with no claim of a significant difference
- The signal is spatial: without fire-neighbourhood features PR AUC drops from 0.248 to 0.048; tuning moves it by about 0.01
- 5% of the training data gives almost the same eval PR AUC (0.204 vs 0.207): the model is limited by bias, not variance
- It finds fire at the front (recall 0.96–0.99 within 1 px) but recall is near 0 beyond 5 km, where 64% of missed pixels lie
- Side finding: the patches are stored south-up, which matters for every wind-based feature

Future work (Phase 2)

- **Whole-patch models:** U-Net / CNN that see the shape of the fire front instead of a fixed-radius halo
- **Imbalance-aware losses:** focal loss or class-weighted segmentation loss
- **Multi-day inputs:** fire history over several days (WildfireSpreadTS)
- **Split leakage check:** whether one fire appears in more than one split (test alone has 40 exact duplicate patches)
- **Uncertain labels:** better handling of cloud and smoke (-1) pixels than simply ignoring them

Thank you

Questions?

Code, report and figures: `github.com/harshilv17/next-day-wildfire-spread`